

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
MASTER OF BUSINESS ADMINISTRATION (MBA)
POST GRADUATE DIPLOMA IN MANAGEMENT (PGDM)
SEMESTER - IV - UNIVERSITY EXAMINATION MAY - 2018
MB829.3 DERIVATIVES AND RISK MANAGEMENT

Date: 19/05/2018, Saturday

Time: 10:00 AM to 01:00 PM

Maximum Marks: 70

Instructions:

1. This paper contains six questions.
2. All questions are compulsory.
3. The first five questions carry 10 marks. The last question carries 20 marks.

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|-----|---|---|----|----|
| Q.1 | A | What do you mean by derivative? Explain how futures contracts differ from forward contracts? | 05 | 10 |
| | B | Discuss each of the following type of traders in a derivatives market- Hedger & Speculator. | 05 | |
| Q.2 | A | What do you understand by commodity futures? What are the benefits of commodity futures at national level? | 05 | 10 |
| | B | As an exporter you expect to receive 3 months from now US \$ 20,000. The spot price of US \$ is Rs 50.00 while 3-m futures at NSE is trading at Rs 49.30 indicating depreciation of US dollar. Under what circumstances would you like to hedge? What would be the hedging strategy? | 05 | |
| Q.3 | A | An investor on March 12, 2010 buys 2000 shares of Infosys at the price of Rs. 390 per share. The portfolio value being Rs. 7,80,000 (Rs. 390x2000). The investor feels that the market will fall and thus needs to hedge by using Infosys Futures (stock futures). What will be his course of action to hedge against possible losses. The Infosys futures (near month) trades at Rs. 402. The Infosys spot price is Rs. 300. | 05 | 10 |
| | B | Calculate the forward price of a non-dividend paying share if, cash price prevails in market is Rs.1200, Futures contract maturity is 1 month from date; assume that market interest rate is 12%. | 05 | |
| Q.4 | | Explain the following and give appropriate example. | | 10 |
| | A | Mark to Market Margin | 05 | |
| | B | Open interest | 05 | |
| Q.5 | A | USD/INR is at 46.46. A trader in Mumbai buys a call option with strike price of 47.00 at a premium of Rs. 0.33. Show his net payoff with different spot prices. (Lot size, 1 contract = 1000 USD). | 05 | 10 |
| | B | If you are bullish on a security and you anticipated that the price of that security will move up within one month time period, you have two choices with you i) You can buy 250 shares of that stock in spot. ii) You can buy one lot of that particular security consists of 250 shares in futures. What will you do? Explain with one example. | 05 | |

- Q.6 A Mr. Anil is bullish about Bajaj Finance Ltd stock. He buys Bajaj Finance Ltd. at 10 20
current market price of Rs. 4000 on 4th July. To protect against fall in the price of
Bajaj Finance Ltd. (his risk), he buys an Bajaj Finance Ltd. Put option with a
strike price Rs. 3900 (OTM) at a premium of Rs. 143.80 expiring on 31st July.
What will be Mr. Anil's Maximum profit and loss? Draw Mr. Anil's payoff
schedule & payoff chart.
- B Suppose an investor Mr. A buys or is holding Eicher Motors Ltd. currently trading 10
at Rs. 4758. He decides to establish a collar by writing a Call of strike price Rs.
5000 for Rs. 39 while simultaneously purchasing a Rs. 4700 strike price Put for
Rs. 27. What will be Mr. A's Maximum profit and loss? Draw Mr. A's payoff
schedule & payoff chart.